Dario Serrano

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Authorized to work in the U.S. (Citizen), EU (Citizen) and Canada (Work Permit)

PROFILE

Mechanical Engineering student at the University of Toronto with experience in mechanical design (SolidWorks, Eagle PCB, ANSYS, Creo, MATLAB) and applied AI (PyTorch). Seeking a 12–16 month co-op/internship (2026) in Mechanical/Mechatronics Engineering, with strong interest in robotics, AI-driven systems and analytical modeling.

EDUCATION

University of Toronto, Toronto, ON — 2023–2027
B.A.Sc. Mechanical Engineering + PEY Co-op
Minor: Robotics & Mechatronics — Certificates: Engineering Business, AI Engineering
Teams: U of T Baja Racing (Suspension) — U of T Racing (Chassis Design)
High school distinctions: International Math Modeling Competition (IMMC) top 3%, Physics Bowl National Gold

ENGINEERING EXPERIENCE

Engineering Intern — Schindler Group — Jul–Aug 2024
Shanghai, China

- Modeled elevator structural components (Ω-Beam, L-Beam) in Creo Parametric with the worldwide R&D team; performed FEA per FKM guidelines.
- Built MATLAB ODE models to analyze system oscillations and validate simulation accuracy.
- 3D-printed lift solution prototypes on industrial systems for rapid design iteration.

Baja SAE Racing — UofT Baja Racing Team (Suspension) — Sep 2025–Present
Toronto, ON

- Contributing to the revival of UofT's Baja SAE program after 6 years of inactivity; preparing an off-road vehicle for the June 2026 SAE competition.
- Performing MIG/TIG welding and manual machining of suspension and chassis components to competition-ready tolerances.
- Designing original suspension parts and frame assemblies in SolidWorks; integrating OTS hardware and components from partner teams into the vehicle architecture.

Androgynous Spacecraft Docking Mechanism — Kinematics & Dynamics — Sep–Dec 2025
Toronto, ON

- Built a parametric SolidWorks model of a 6-arm variable-angle rough-alignment mechanism for a microsatellite docking interface.
- Wrote MATLAB kinematic/static analysis code; solved a 6×6 equilibrium system for joint reactions under a 6 kN axial load and determined a required slider actuator force of 2.17 kN via Jacobian-based force mapping.

Current Sensing & Anti-Aliasing PCB — Digital and Analog Circuits — Jan–May 2026
Toronto, ON

- Designed an analog current-sensing front-end (LM324, 0–3.5 A to 0.25–3.3 V) and a 3rd-order Chebyshev LPF (1 kHz cutoff, ≥35 dB at 5 kHz) for 50–70 kHz ADC sampling; validated in LTspice.
- Laid out a 2-layer PCB in EAGLE with controlled power/ground routing; generated Gerber files and hand-soldered the prototype.

CNC Mill Design — Mechanical Engineering Design — Sep–Dec 2024
Toronto, ON

- Designed a market-ready desktop CNC mill in SolidWorks; performed drive trade studies (stepper vs servo) and cost–performance analysis for manufacturability.

AI & ANALYTICAL PROJECTS

Seizure Detection with CNN–LSTM — Applied Fundamentals of Deep Learning — May–Sep 2025
Toronto, ON

- Built a CNN–LSTM in PyTorch for EEG seizure classification (CHB-MIT dataset); implemented Butterworth filtering, wavelet denoising, and sliding-window segmentation.
- Achieved AUROC 0.96 and F1-score 0.84 on unseen patients, demonstrating strong cross-patient generalization.

Research on Rocket Optimization — Jun–Oct 2022
Remote

- Published "Applications of Optimization Techniques for Solid Rocket Design" (Darcy & Roy Press, DOI: 10.54097/hset.v38i.5936); applied numerical optimization methods for propulsion performance.

TECHNICAL SKILLS

Design & Fabrication: SolidWorks, Creo, ANSYS, 3D Printing, Machining, MIG/TIG Welding
Programming: Python, PyTorch, MATLAB, NumPy, pandas, SciPy
Other: Microsoft Office, Minitab   Languages (Native): English, Mandarin, Shanghainese

Reference: Available upon request from Daryoush Ziai, CEO — Schindler Asia Pacific.